Reproducibility and Provenance

Reproducibility and Provenance I

The reproducibility contract is:

Reproducibility and Provenance I

1. construct the same frozen parameter dataclass;
2. use the same generator ID and the same deterministic seed;
3. compare the canonical artifact digest;
4. record the parameter schema, typed parameters, taxonomy, evidence boundary, canonical byte serialization, objective metrics, and encoding request;
5. if a file is written, inspect the decoded media and compare dimensions, channels, rates, frame counts, durations, timing offsets, and encoded hash;
6. retain the v2 manifest and source-data sidecars with the generated figure or table. Note that in this release the generators are analytic in their typed parameters: the request seed s is recorded in manifests and study plans and excluded from the canonical digest, but it does not yet alter artifact bytes; seed-driven stimulus randomization awaits a stochastic generator.

The contract is deliberately stronger than “the file can be downloaded.” Research-software guidance treats versioned code, executable procedures, dependencies, and retained inputs as part of the reproducible object (Sandve et al. 2013; Wilson et al. 2014). FAIR guidance further asks that digital research objects be findable, accessible, interoperable, and reusable, while noting that software has lifecycle and maintenance constraints that are not identical to those of static data (Wilkinson et al. 2016; Lamprecht et al. 2020). DuckRabbit operationalizes those principles with typed metadata, deterministic hashes, source-data sidecars, explicit licenses, and release gates rather than by implying that a generated media file is a complete scientific replication.

Reproducibility and Provenance I

The package reports implemented, planned, input_required as its catalog status vocabulary and currently covers audio-visual, auditory, visual. The evidence layer contains 18 entry records backed by 36 source records, covering 18/18 catalog entries. The publication workflow produces 15 figures and 10 tables from the live registry.

The public software record is the DuckRabbit public GitHub repository, with Daniel Ari Friedman of the Active Inference Institute as author. The release is archived on Zenodo at 10.5281/zenodo.21419693 (concept DOI, always resolves to the latest version), with 10.5281/zenodo.21419694 identifying this v0.5.0 release specifically.

Reproducibility and Provenance I

The canonical buffer is the primary identity of a stimulus. Encoded files are delivery artifacts with format-specific tolerances. The observer protocol is a separate evidence layer: a manifest can prove what was presented without proving what an observer experienced.

Synthetic psychophysics is a third, deliberately bounded object: the model is a hand-specified deterministic feature observer; no training data; human_data=false, registered as `duckrabbit.synthetic.feature_observer`. The model version, feature schema, weights, temperature, seed, calibration statement, canonical reference/comparison digests, and `human_data=false` flag are retained in `output/data/synthetic_psychophysics.json`. Re-running this diagnostic checks deterministic model orchestration and stimulus-feature sensitivity; it does not estimate a human observer or replace empirical psychophysics.

Reproducibility and Provenance I

The v2 verifier is a relational check rather than a field-presence check. It rejects summaries whose dimensions, dtype, byte count, rates, duration, signal range, frame deltas, or signed audiovisual offset disagree. Public manifest and inspection mappings are recursively frozen after validation, and rational frame rates are reduced before entering the clock contract. The read-only capability probe records whether Pillow, NumPy, ffmpeg, or ffprobe are available in the current environment; unavailable delivery paths remain explicit rather than being silently downgraded.

The minimal regeneration commands are:

```
uv run pytest -q --cov=src/duckrabbit --cov-fail-under=90
uv run python scripts/generate_publication_outputs.py
uv run python scripts/generate_manuscript_variables.py
```

No participant records, fitted observer coefficients, or human psychophysics are bundled. The synthetic diagnostic is explicitly model output with `human_data=false` and `training_data=none`. Reuse should cite Daniel Ari Friedman and the release metadata in `CITATION.cff`, together with the source-specific scholarship in `references.bib`. Software-citation principles recommend citing the software object itself, with enough version and identity information to distinguish one release from another (Smith et al. 2016). The DOI field carries the real, minted concept DOI (`10.5281/zenodo.21419693`) and `doi_status` is `published`; no resolver URL was manufactured before that identifier existed.